

Least-cost electrification pathways for Senegal by 2030: A nationwide analysis using open-source spatial electrification tool (OnSSET)

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ABSTRACT

Achieving universal electricity access in Senegal by 2030 remains a major policy challenge due to persistent spatial disparities in infrastructure, population density, and resource availability. This study conducts a nationwide, spatially explicit assessment of least-cost electrification pathways using OnSSET. The analysis develops context-specific scenarios to plan optimal technology mixes across rural and peri-urban areas, based on differentiated tiers of electricity access. By integrating high-resolution geospatial, demographic, and techno-economic data, the model identifies the most economically viable solutions for achieving universal access. Results indicate that grid extension is the least-cost option for approximately 93.7 % of the population, largely concentrated in peri-urban areas with high population density and proximity to existing grid infrastructure. In contrast, solar PV mini-grids (MG PV) and stand-alone PV (SA PV) systems are optimal for 0.7 % and 5.6 % of the population, respectively, mainly in remote, sparsely populated rural settlements. The total investment required to achieve universal electricity access by 2030 is estimated at USD 269.8 million, corresponding to 116.1 MW of additional installed capacity.

Beyond quantifying cost-optimal solutions, the study demonstrates the potential of open-source geospatial models like OnSSET to support transparent, data-driven planning in developing country contexts. It also highlights key policy implications, emphasizing the need for integrated national electrification strategies that combine centralized and decentralized systems to address regional disparities. Limitations of the study include uncertainties in input data quality, static demand assumptions, and the exclusion of non-technical barriers such as institutional capacity and financing constraints. Nonetheless, the findings provide a valuable decision-support basis for Senegal's ongoing energy transition and broader Sustainable Development Goal 7 (SDG7) objectives.

Introduction

Access to reliable and affordable electricity is a key driver of socio-economic development, particularly in developing countries. However, >30 % of rural communities remain unconnected to the national grid, highlighting persistent disparities between urban and rural areas [1]. This disparity is particularly pronounced in Sub-Saharan Africa (SSA) [2]. In 2023, approximately 600 million individuals across SSA remain without constant and consistent access to electricity, representing almost half of the continent's population and accounting for over 80 percent of the global electricity access deficit [3,4].

In Sub-Saharan Africa, electricity generation is predominantly based on fossil fuels and hydropower, with natural gas accounting for 46 TWh,

oil for 45 TWh, and hydropower for 23 TWh [5]. In 2022, Senegal electric energy production amounts to 1826 kilotonnes of oil equivalent (ktoe), with renewable energy accounting for only 14 % of electricity generation. The total primary energy supply reaches 5523 ktoe. The household electricity access rate is estimated at 71 %, while the average electricity consumption per capita stands at 409 kWh/person [6]. The Senegalese government has committed to achieving universal electricity access by 2030, aligning with Sustainable Development Goal 7, which aims to ensure access to affordable, reliable, sustainable, and modern energy for all [7,8]. Several studies have supported rural electrification planning in developing countries through various modeling and data-driven approaches [9] developed model-based scenarios to assess long-term electrification pathways, emphasizing their relevance for

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shaping energy policy in rural contexts. In a case study focusing on Myanmar, Pote et al. [10] proposed sustainable solutions for rural electrification by highlighting the role of renewable energy technologies and context-specific financial mechanisms. Geographic Information Systems (GIS) have also been utilized to support electrification strategies, as demonstrated by Taye et al. [11] who applied a GIS-based methodology to identify cost-effective electrification options. Additionally, Girona et al. [12] introduced an interactive planning tool that integrates geospatial and socio-economic data to support universal electricity access efforts. Some approaches rely on the deployment of mini-grid systems, which enable decentralized electricity distribution to isolated or underserved communities beyond the reach of the national grid [13–15]. These systems, often powered by renewable energy sources such as solar or hydropower, provide an efficient solution to meet local energy demands while reducing reliance on fossil fuels.

More recent approaches have leveraged Geographic Information Systems (GIS) as a central tool for planning the electrification of remote sites [16–18] proposed the development of an enterprise-level GIS integrated with smart grid infrastructure to enhance the management and planning of modern energy systems. By integrating various spatial data, such as population density, accessibility to the existing power grid, and local energy resources, these methods optimize the selection of technologies and infrastructure required for effective and sustainable electrification [19–22]. The use of data from GIS provides an effective solution to address the data scarcity often observed in developing countries. By leveraging these technologies, it is possible not only to enhance the availability and accuracy of information but also to significantly reduce the time required for electrification project planning [23]. The SolarGIS method is utilized for rural electrification planning by leveraging its high-resolution satellite data on solar irradiation and energy resources [24–26]. Nolan et al. [27] employed the Network Planner model to optimize rural electrification by minimizing installation costs for mini-grids in developing countries. The Network Planner model has also been applied in other studies for electricity access planning in various countries [28,29]. In certain studies, the reference electrification model has been deployed to identify the most cost-effective electrification options for each individual consumer [30–34]. Recent studies utilize the OnSSET model to conduct a more comprehensive and in-depth analysis of rural electrification options, considering costs, existing infrastructure, and renewable energy potential [35–41]. In Senegal, Sanoh et al. [42] used the Network Planner model to assess electrification planning, highlighting key cost drivers and policy implications. Diouf and Miezan [43] proposed a cost-effective model for universal access in small villages in Senegal, highlighting its potential to achieve full rural coverage at a significantly reduced cost. Bold [44] analyzes the impact of electricity sector reforms in Senegal on capital dynamics, power relations, and the potential for a fair transition to a low-carbon energy system. For instance, previous studies in Senegal have demonstrated that optimizing the tilt angle of PV panels significantly influences energy yield, highlighting the need to account for local climatic and solar conditions when planning solar-based electrification solutions [45].

OnSSET enables the simulation of various electrification scenarios tailored to local contexts. In Ethiopia, Temesgen et al. [46] employed the model to analyze viable grid extension options. Bissiri et al. [47] underscored the critical need to integrate spatial and temporal variability of variable renewable energy sources into regional capacity expansion planning in West Africa. Numerous African countries have incorporated this model into their strategic frameworks to achieve universal electricity access [48–52]. Jean et al. [53] used the OnSSET model to identify cost-effective electrification options for a municipality in the Brazilian Amazon, highlighting solar energy as the most suitable solution. Pueyo et al. [54] highlighted the limitations of traditional power system planning in SSA, advocating for models that better reflect the realities of both grid and off-grid development. Dagnachew et al. [55] proposed a methodology, based on the image-timer framework, to

assess how productive uses of electricity influence regional power systems. A geospatial database and modeling tool have been developed to help planners in Afghanistan identify and visualize the most cost-effective electrification pathways to achieve national electricity access goals [56]. The model has been applied at broader scales to evaluate electrification strategies within regional planning frameworks. Thus, Mentis et al. [57] developed and compared multiple electrification scenarios to identify the most cost-effective options for expanding electricity access in SSA. Similarly, Bertheau et al. [58] analyze electrification options in SSA using geospatial data and scenarios to identify the most suitable solutions. Sayani et al. [59] utilized publicly available energy modeling tools, integrated with geospatial data, to determine the most cost-effective power generation investment strategies across 48 African countries. In the same way, [60] provided a comprehensive review of electricity planning and implementation research across all SSA countries, regional power pools, and the broader subcontinent. Bissiri et al. [61] also investigated, using the OnSSET tool, the optimal mix of grid-based and off-grid electrification solutions to meet rural and urban energy demand in Burkina Faso and Côte d'Ivoire by 2030. The literature also presents alternative approaches to energy systems planning [62–66].

Several studies have addressed energy planning in SSA, particularly to identify least-cost electrification options under technical, economic, and geographic constraints. However, most of these efforts focus on regional or multi-country scales, with limited application to the specific context of Senegal.

In the Senegalese context, there is a notable lack of studies aiming to achieve universal electricity access through the integration of renewable energy while prioritizing economically optimal solutions. This gap highlights the relevance of our contribution, which seeks to address this shortcoming through a detailed and context-specific analysis.

Based on this comparison Table 1, OnSSET was selected for this study due to its ability to incorporate spatially explicit data, compare multiple technology options, and provide flexible scenario development capabilities. Its open-source nature also facilitates transparency and reproducibility. Moreover, the model has been applied in various national electrification planning processes, which further supports its credibility

Table 1
Comparison of selected electrification planning models.

Model	Description	Strengths	Limitations
OnSSET	A geospatial, open-source, Python-based model designed for least-cost electrification planning.	Integrates spatial data, allows comparison of grid, mini-grid, and standalone systems, customizable, supports scenario analysis.	Requires detailed geospatial input, some familiarity with Python and QGIS needed.
Electrification Pathways Tool (EPT)	Developed by the World Bank for quick least-cost scenario evaluation.	User-friendly interface, suitable for rapid assessments.	Limited flexibility for customized analysis, less transparent in terms of underlying assumptions.
Network Planner	A tool from Columbia University focused on distribution network expansion planning.	Well suited for grid extension analysis, incorporates network topology.	Less focus on off-grid or mini-grid alternatives, less suitable for dispersed rural populations.
The Renewable Energy Model (TREM)	A techno-economic optimization tool for renewable energy planning in power systems.	Effective for analyzing renewable energy integration, good for grid-connected systems.	Not well suited for rural electrification or off-grid solutions, requires detailed techno-economic inputs.

and practical relevance. These features make OnSSET particularly suitable for assessing least-cost electrification pathways across diverse geographic and socio-economic contexts, such as in Senegal.

Methodology

In this study an adapted version of OnSSET to the national context of Senegal was used, with a specific focus on rural and peri-urban areas. OnSSET is a geospatially explicit modeling framework designed to support least-cost electrification planning by evaluating multiple technological options such as grid extension, mini-grids, and stand-alone systems based on the levelized cost of electricity (LCOE). While the tool has been widely applied in global and regional studies, this research introduces several context-specific adaptations to enhance the relevance and reliability of the results for national policy planning in Senegal.

Unlike default implementations, this study configures OnSSET using locally adapted techno-economic and demand assumptions to better reflect the Senegalese context. Electricity demand is estimated following the World Bank's Multi-Tier Framework (MTF) [67], which defines access levels according to annual per capita consumption and the range of appliances that can be supported. In this analysis, Tier 2 demand is assigned to rural settlements, covering basic services such as lighting, phone charging, and small appliances. Tier 3 demand is applied to peri-urban areas, where medium-power appliances such as televisions, fans, and refrigerators are common. These differentiated tiers guide both system sizing and technology selection based on settlement characteristics.

In addition to demand tiers, the simulation integrates technology-specific cost, operation and maintenance (O&M), and lifetime parameters derived from recent national and regional benchmarks [68–71]. The discount rate is set at 10 %, while technology lifetimes vary between 10 and 30 years depending on the system type. 10 years for SA diesel, 15 years for SA PV and MG diesel systems, 20 years for MG PV and wind systems, and 30 years for centralized grid and mini-hydro technologies. Corresponding annual O&M costs are expressed as a percentage of the total investment cost: 10 % for centralized grid and diesel-based systems, and 2 % for MG hydro, MG wind, MG PV, and SA PV systems. These techno-economic assumptions directly influence the levelized cost of electricity (LCOE) calculations, ensuring that the modeled outcomes reflect realistic long-term investment, operation, and system performance dynamics under Senegalese conditions.

The geospatial inputs are prepared through extensive preprocessing using QGIS [72], including the disaggregation and cleaning of population data, refinement of electrification status from satellite-derived nighttime lights, and mapping of existing infrastructure such as MV and HV lines. Renewable energy sources such as solar irradiance and wind capacity factors alongside diesel fuel availability are integrated using temporally and spatially resolved datasets specific to Senegal. This enables more accurate assessment of the technical performance and economic viability of PV, wind, and diesel-based systems. Geographic constraints such as road accessibility and terrain are also considered when evaluating technical and economic feasibility.

In addition, technology allocation within the model incorporates spatial and technical constraints, including maximum allowable grid extension distances, minimum capacity factor thresholds for renewables (e.g., solar and wind), and constraints linked to fuel logistics in remote areas for diesel-based systems. These constraints reflect both technical and institutional realities in Senegal and differentiate the model from purely theoretical or default applications.

The model is implemented in Python within a Jupyter Notebook environment [73], where scenario simulations are conducted to evaluate least-cost electrification outcomes. Outputs include the spatial allocation of electrification technologies, estimates of new generation capacity, and associated investment requirements. These outputs are visualized using QGIS to generate maps and summaries that support policy-relevant insights.

Overall, the adapted methodology enables the development of a robust, reproducible framework tailored to national priorities. It offers a practical decision-support tool for stakeholders involved in planning electricity access under realistic constraints of geography, infrastructure, and investment capacity.

Data collection for OnSSET

OnSSET enables the identification of least-cost electrification strategies at the settlement level by leveraging high-resolution geospatial data. Through a techno-economic assessment, it minimizes the total cost of electrification for each settlement over a multi-year planning horizon, and selects the most appropriate technology such as grid extension, mini-grid, or stand-alone system based on local demand, distance to infrastructure, and resource availability [74]. Aiming to ensure full access to affordable, reliable, sustainable and modern electricity for all by 2030. The model requires a comprehensive dataset covering geographical, demographic, technical, and economic parameters to identify the most cost-effective electrification solutions [75]. Key data inputs include population distribution, existing electricity infrastructure, renewable energy potential, road networks, and economic indicators [76,77]. These datasets are sourced from national surveys [78], government databases, satellite imagery, and international repositories such as the World Bank [79], IEA [80], and the Global Solar Atlas [81], among others. Such comprehensive data integration ensures that the model can produce spatially differentiated electrification strategies aligned with local needs and resource availability. Table 2 below summarizes the essential data required for OnSSET.

Table 3 presents essential non-spatial data required for electrification planning. These parameters, such as demographic trends, technology costs, and policy frameworks, help optimize least-cost electrification strategies in underserved regions.

Key metric using OnSSET: LCOE

The LCOE calculation helps compare different electrification technologies by considering all relevant costs over the lifetime of the system and identifying the most cost-effective option for rural electrification. The formula for LCOE is given by:

$$LCOE = \frac{\sum_{t=0}^{n-1} \frac{I_t + O\&M_t + F_t}{(1+r)^t}}{\sum_{t=0}^{n-1} \frac{E_t}{(1+r)^t}}$$

I_t : Investment expenditures for a specific system in year t

$O\&M_t$: the operation and maintenance costs

F_t : the fuel expenditures

E_t : the generated electricity

Table 2
Identification of required GIS data.

Dataset	Type
Administrative boundaries	Polygon vector
Population	Polygon vector
Nighttime light	Raster
GHI	Raster
Wind speed	Raster
Travel hours	Raster
Elevation	Raster
Land cover	Raster
Custom Demand	Raster
Existing HV lines	Lines vector
Planned HV lines	Lines vector
Existing MV lines	Lines vector
Planned MV lines	Lines vector
Substations	Points vector
Roads	Lines vector
Hydropower	Points vector

Table 3

Non-geospatial data.

Electrification rate	Population growth	Technology cost	Demand level and target	Policies
Urbanization rate	Household size	Technical specifications	Fuel prices	Other characteristics

 r : the discount rate n : the lifetime of the system*Techno-economic parameters*

All techno-economic assumptions used for capital cost, operation and maintenance, and LCOE estimation were based on data provided in recent regional reports [82–86]. These sources provide realistic benchmarks for West Africa and the Senegalese context, and were used to parameterize the OnSSET model accordingly.

Application context: Senegal

Fig. 1 presents a contextual map of Senegal. Senegal is administratively organized into 14 regions, which are the highest level of government administration. Each region is overseen by a regional council whose members are elected through general elections. Further subdivisions break these regions down into 46 departments [87].

Senegal features three main agroclimatic zones that shape both settlement patterns and renewable energy potential. The semi-desert zone in the north is dry and sunny, offering high potential for solar energy but limited hydropower resources. The wooded savannah in the center has moderate rainfall and supports agriculture and livestock, making it suitable for solar and biomass systems. In contrast, the forested zone in the south experiences higher rainfall and denser vegetation, providing favorable conditions for the deployment of hybrid renewable energy systems.

Senegal, located in West Africa, has a diverse energy landscape characterized by a mix of grid-connected and off-grid electrification solutions. Despite significant progress in expanding electricity access, rural areas still face challenges due to their remoteness and limited

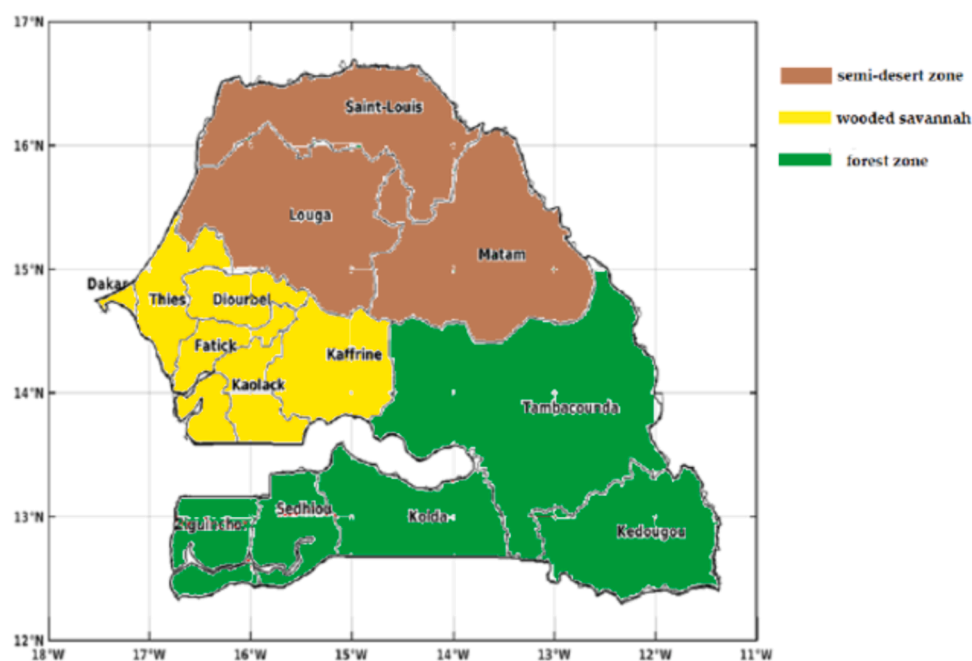
infrastructure. The country's energy strategy prioritizes renewable energy integration, with a strong focus on solar photovoltaics, to achieve universal electrification. The application of OnSSET in Senegal helps identify the most cost-effective pathways to expand electricity access, particularly in rural and underserved regions. Rural development is closely linked to access to electricity. Reliable power supply enables food preservation and cold storage, which are essential for maintaining the quality of agricultural and dairy products. This, in turn, supports the establishment of local value chains and facilitates the distribution of these products to urban and export markets.

Fig. 2 illustrates the spatial distribution of rural and urban areas within the study region. In the visualization, red points represent urban areas, while blue points indicate rural areas. A key observation from the spatial distribution is the predominant concentration of urban centers in the western part of the country, suggesting a higher population density in this region.

The classification of settlements into rural and urban categories in this study is based on the average household sizes reported by the Agence Nationale de la Statistique et de la Démographie (ANSD) [89] during the census. Settlements were classified as rural or urban based on average household size, following ANSD [90], which reports 7.6 persons per household in urban areas and 10.8 in rural areas. This approach provides a practical and consistent proxy for demographic differences across settlements, which is essential for estimating energy demand and determining least-cost electrification options in OnSSET.

Table 4 below presents settlement characteristics in Senegal based on OnSSET for the base year 2018 and the projected year 2030. The data provide insights into demographic changes, population distribution, and their implications for electrification planning. The total number of settlements remains at 58,701, with 58,542 rural settlements and 159 urban settlements. The minimum population per settlement increases slightly from 6 to 7 inhabitants, suggesting a marginal rise in the least populated areas, but also suggesting that some areas are inhabited by single families not living close to others. The maximum population per settlement experiences a significant increase from 543,108 to 803,650, reflecting rapid urbanization and population concentration in major urban centers, with a growth of 260,542 inhabitants in the most populated locality.

In this study, the spatial distribution of settlements is derived from a

**Fig. 1.** Contextual map of Senegal by administrative and agroclimatic zones [88].

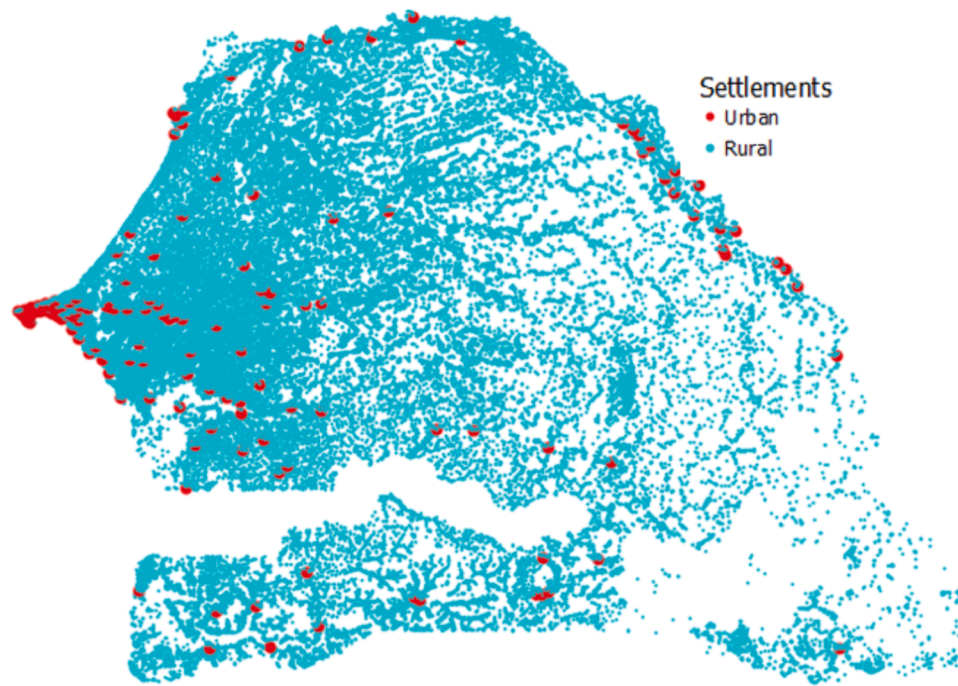


Fig. 2. Urban and rural settlements.

Table 4
Senegal population in the base year 2018 and the end year 2030.

	Population 2018	Population 2030
Total	15,726,037	21,734,270
Urban population	7344,020	10,867,135
Rural population	8382,017	10,867,135

georeferenced population raster layer, corresponding to the baseline year (2018). The number of settlements (58,701) remains constant throughout the planning horizon, as OnSSET uses a fixed spatial reference to ensure consistency in geospatial analysis. However, the population associated with each settlement is updated over time using national projections provided by the Agence Nationale de la Statistique et de la Démographie [90]. This approach allows for temporal evolution of population data while preserving the spatial granularity of the initial dataset. As a result, a small number of urban settlements 159 out of 58,701 (Table 5) concentrates approximately 46.70 % of the national population in 2018, reflecting the demographic reality of Senegal where large cities have attracted a significant share of the population. This method enables the model to capture urbanization trends and

Table 5
Senegal settlements characteristics in the base year 2018 and the end year 2030.

	Base year	Step year	Difference
Number of settlements	58,701	58,701	–
Rural settlements	58,542	58,542	–
Urban settlements	159	159	–
Minimum population of settlement	6	7	1
Maximum population of settlement	543,108	803,650	260,542
Number of settlements with a population of 0–100	44,624	41,832	–2792
Number of settlements with a population of 101–500	10,355	11,966	1611
Number of settlements with a population of 501–1000	2036	2511	475
Number of settlements with a population above 1000	1678	2300	622

demographic growth without modifying the geospatial settlement framework.

Table 5 shows distribution of settlements across different population size brackets highlights notable demographic shifts. The number of small settlements (0–100 inhabitants) decreases by 2792, implying that many small villages are experiencing population growth and transitioning into higher population categories. The number of settlements in the 101–500 population range increases by 1611, while those in the 501–1000 range increase by 475, indicating a demographic shift towards medium-sized rural settlements. The number of settlements exceeding 1000 inhabitants grows from 1678 to 2300, reflecting an increase of 622 settlements, which signals rural densification and the expansion of peri-urban areas.

Nighttime light

Fig. 3 illustrates nighttime light emissions across Senegal, with red points indicating electrified areas and black points corresponding to non-electrified settlements. This spatial distribution aligns closely with known patterns of population density, urbanization, and existing grid infrastructure. The most intense concentrations of nighttime lights are observed in and around Dakar and the Cap-Vert Peninsula, which correspond to the most urbanized and densely populated part of the country. This region benefits from well-developed grid infrastructure, high electricity demand, and significant economic activity, all of which contribute to greater light emissions detectable by satellite.

Moving away from urban centers, the frequency of nighttime lights decreases markedly, especially in the northeastern and southeastern regions. These areas are characterized by sparser populations, limited economic activity, and in many cases, infrastructure deficits, which hinder grid extension. The distance from the existing medium- and high-voltage transmission lines also plays a role, as electrification in remote settlements often becomes economically prohibitive without targeted investments or alternative decentralized solutions such as mini-grids or stand-alone PV systems.

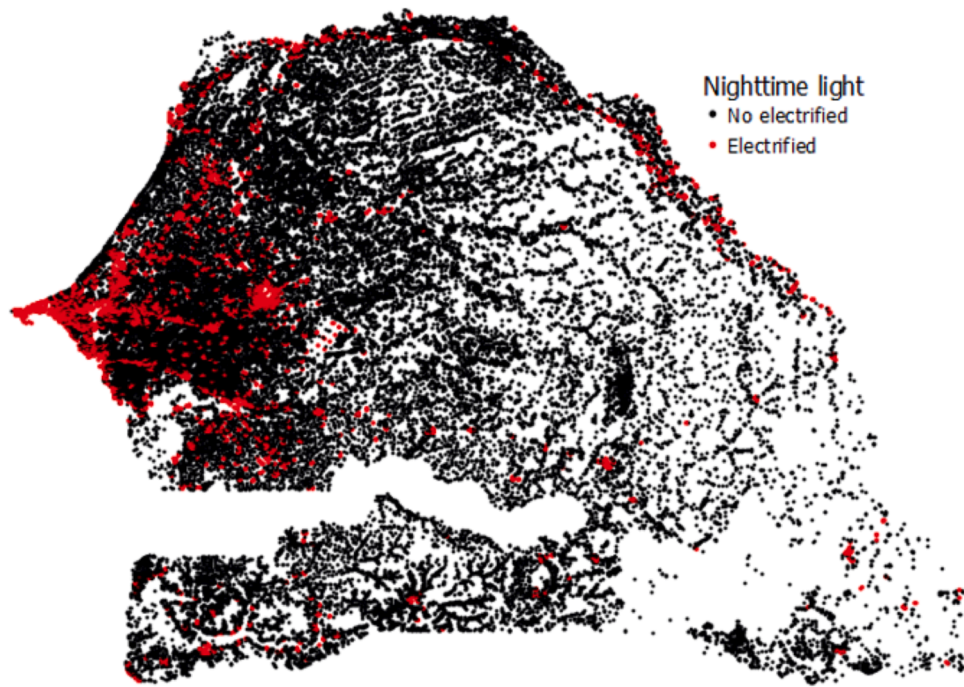


Fig. 3. Senegal nighttime lights.

Electricity transmission network

The absence of nighttime light in many areas, despite human settlement, reinforces the need for targeted rural electrification strategies and highlights the gaps that models such as OnSSET aim to address by identifying least-cost solutions based on spatial, demographic, and economic data.

Fig. 4 complements the spatial analysis of population distribution

and nighttime light emissions by presenting a map of the Senegal electricity transmission network, which illustrates the geographic extent of existing high-voltage and medium-voltage infrastructure. The existing high-voltage (HV) transmission network in Senegal is operated by SENELEC and the regional utility OMVS/SOGEM. It includes a 225 kV international interconnection linking the Manantali hydropower plant in Mali to Dakar via the northern corridor through Mauritania. This main axis extends toward the Touba and Kahone substations, facilitating

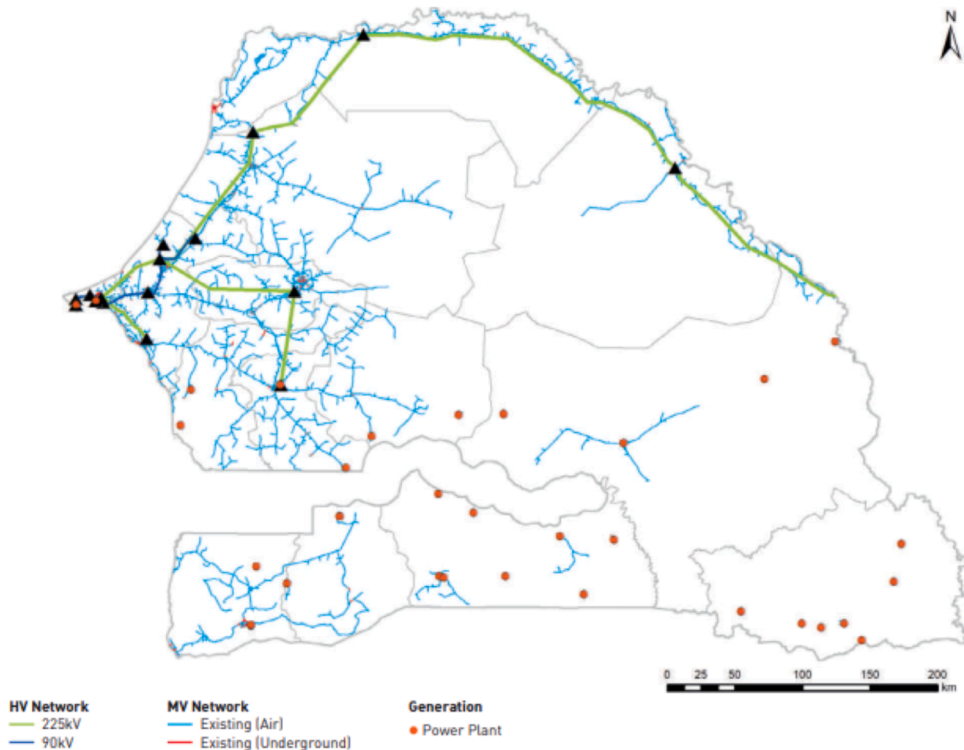


Fig. 4. Senegal electricity transmission network HV and MV power lines [91].

both the evacuation of generation from plants in the central region and the supply of electricity to the regions of Diourbel, Kaolack, Kaffrine, and Fatick. The HV network also comprises 90 kV and 225 kV lines serving the main urban centers of Dakar, Thiès, and Mbour. Overall, the interconnected grid covers most of northern and western Senegal, linking major cities and towns. In contrast, the southern regions such as Ziguinchor, Kolda, Tambacounda, and Velingara are supplied by several autonomous medium-voltage mini-grids that form part of the non-interconnected system.

Results and discussion

Electricity demand and additional capacity requirements

The projected electricity demand associated with new connections exhibits considerable variation across the study area, ranging from 0.592 MWh to 17,257 MWh per settlement, with an average of 7.295 MWh. This demand estimate corresponds exclusively to household-level consumption and does not include productive or institutional uses. The variation reflects differences in population size, household density, and the assigned service tier. Larger and more urbanized areas tend to exhibit higher total domestic energy needs, in line with higher per capita consumption assumptions. In contrast, smaller rural settlements present more modest consumption profiles under Tier 2 standards, often limited to basic lighting and appliance use [Fig. 5](#).

To satisfy this domestic demand, the model estimates the installed generation capacity required for each locality based on the least-cost electrification technology. This required capacity ranges from 0.01 kW to 2938.53 kW, with an average of 1.98 kW per locality (see [Fig. 6](#)). The variation is influenced not only by the scale of domestic demand but also by the performance characteristics of the selected energy systems. Accurate estimation of household consumption is thus essential to ensure adequate sizing, technical feasibility, and cost-effectiveness of electrification pathways.

Minimum overall technologies

With reference to [Fig. 7](#), the model aims to identify the most

economically viable electrification options to achieve universal electricity access by 2030. The model evaluates multiple technologies, including electric grid extension, mini-grid systems (incorporating photovoltaic (MG PV), wind, hydropower, and diesel generation), and stand-alone systems (SA PV and SA diesel).

In the context of the Senegalese case study analyzed using the OnSSET model, three economically viable electrification pathways are identified: grid extension (represented in blue), solar PV-based mini-grids (in yellow), and stand-alone solar PV systems (in green).

Wind-based MG were not selected as cost-optimal solutions for rural or urban electrification. This outcome can be directly linked to the limited wind energy potential observed across most regions of Senegal. Specifically, the wind capacity factor reaches a maximum of only 20.63 % in a few select locations. However, for the majority of settlements, the capacity factor remains below 10 %, with a minimum as low as 0.3 %. Such low-capacity factors significantly reduce the energy yield of wind systems, thereby increasing their LCOE. As OnSSET selected technologies based on least-cost criteria, wind mini-grids are systematically outperformed by solar-based solutions, which benefit from more consistent and abundant solar irradiance across the country. Moreover, the intermittency and spatial limitation of wind resources make them less attractive for decentralized electrification planning in the Senegalese context.

Hydro-based MG were also not selected as cost-effective options. Although Senegal has some hydro potential, it is geographically constrained to specific regions, primarily near major rivers such as the Senegal River, where it has already been developed significantly. Most rural settlements are located far from these resources, making the development of small hydro plants technically challenging and economically unattractive due to the need for extensive transmission infrastructure. Additionally, hydrological variability and seasonality can affect the reliability of supply, especially for small-scale run-of-river systems. These factors collectively increase both the technical risk and the LCOE, reducing the competitiveness of hydro-based MG in comparison to other options.

Diesel-based solutions, in both MG and SA systems, were also not selected as least-cost options in the model. This is largely attributed to the high and volatile cost of diesel fuel in Senegal, particularly in remote

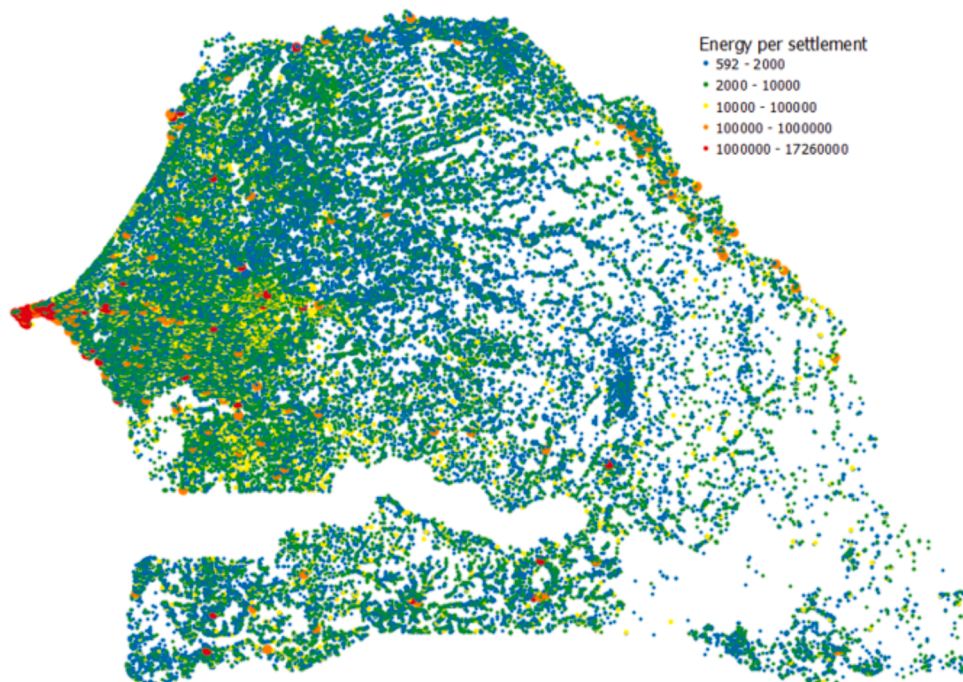


Fig. 5. Annual electricity demand in 2030 (kWh) based on new connections.

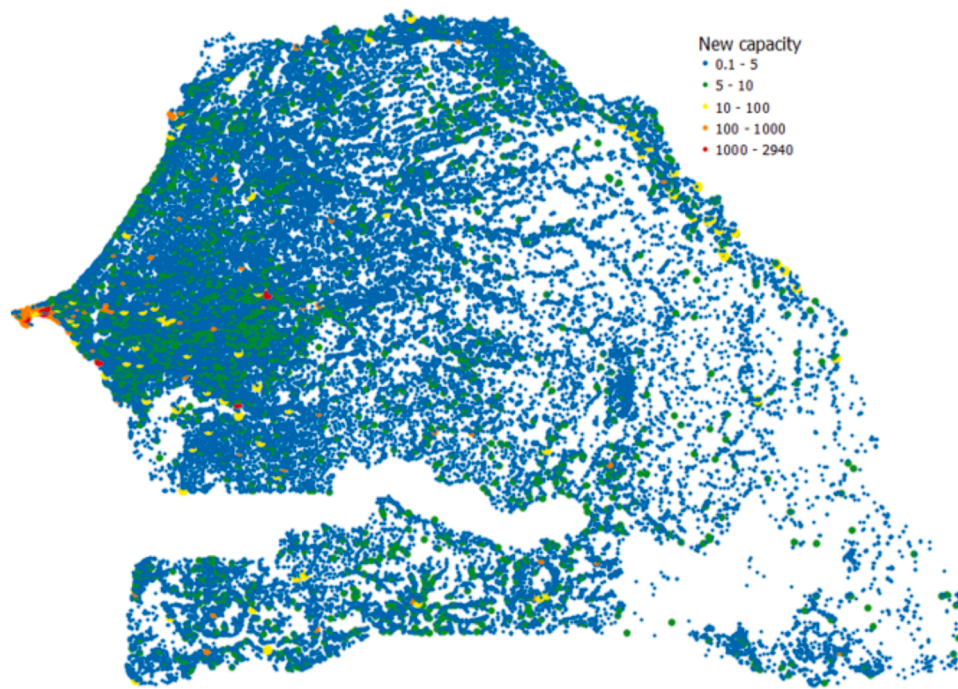


Fig. 6. Additional capacity (kW).

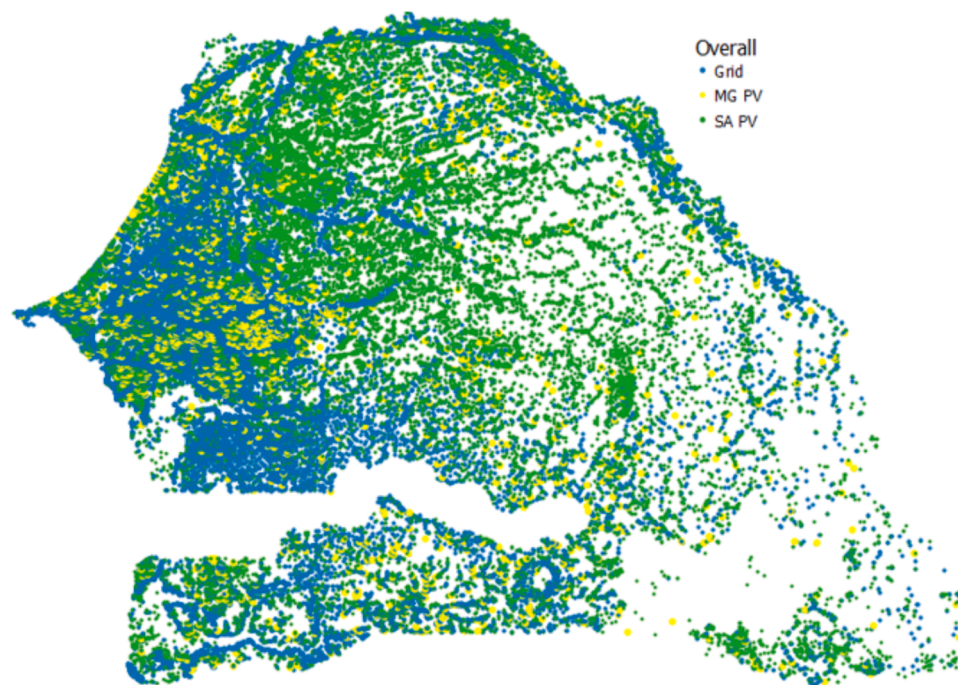


Fig. 7. Type of technology providing electricity in 2030.

areas where transportation costs can substantially affect the price at the service station. Furthermore, operational and maintenance costs for diesel generators remain relatively high, and their lifespan is generally shorter compared to renewable alternatives. Beyond cost, diesel technologies are increasingly disfavored due to their environmental impact, and many governments, including Senegal's, are adopting policies that limit new diesel-based electrification projects. Consequently, diesel options are consistently outperformed by solar PV-based solutions, which have benefited from substantial cost declines and are more scalable in off-grid contexts.

These results confirm a broader trend observed across SSA, where solar PV either as MG or SA systems has emerged as the dominant least-cost option for rural and peri-urban electrification. Wind, hydro, and diesel-based systems can play a role in specific localized contexts, but their general techno-economic competitiveness remains limited under national-scale optimization approaches such as OnSSET.

Off-grid technologies

Fig. 8 illustrates the distribution of different off-grid electrification

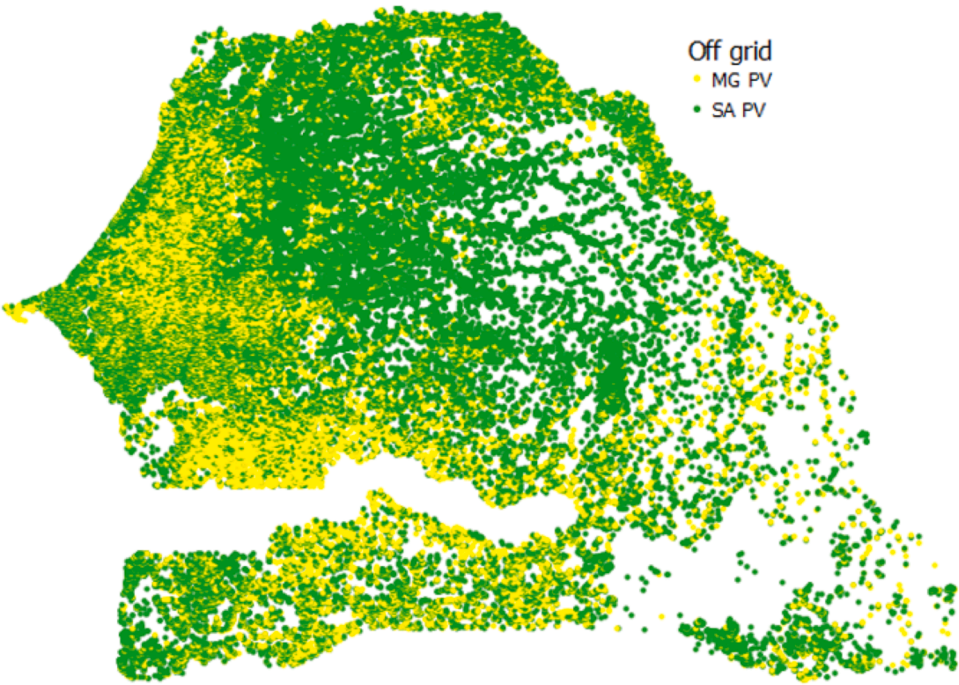


Fig. 8. Type of off-grid technology providing electricity in 2030.

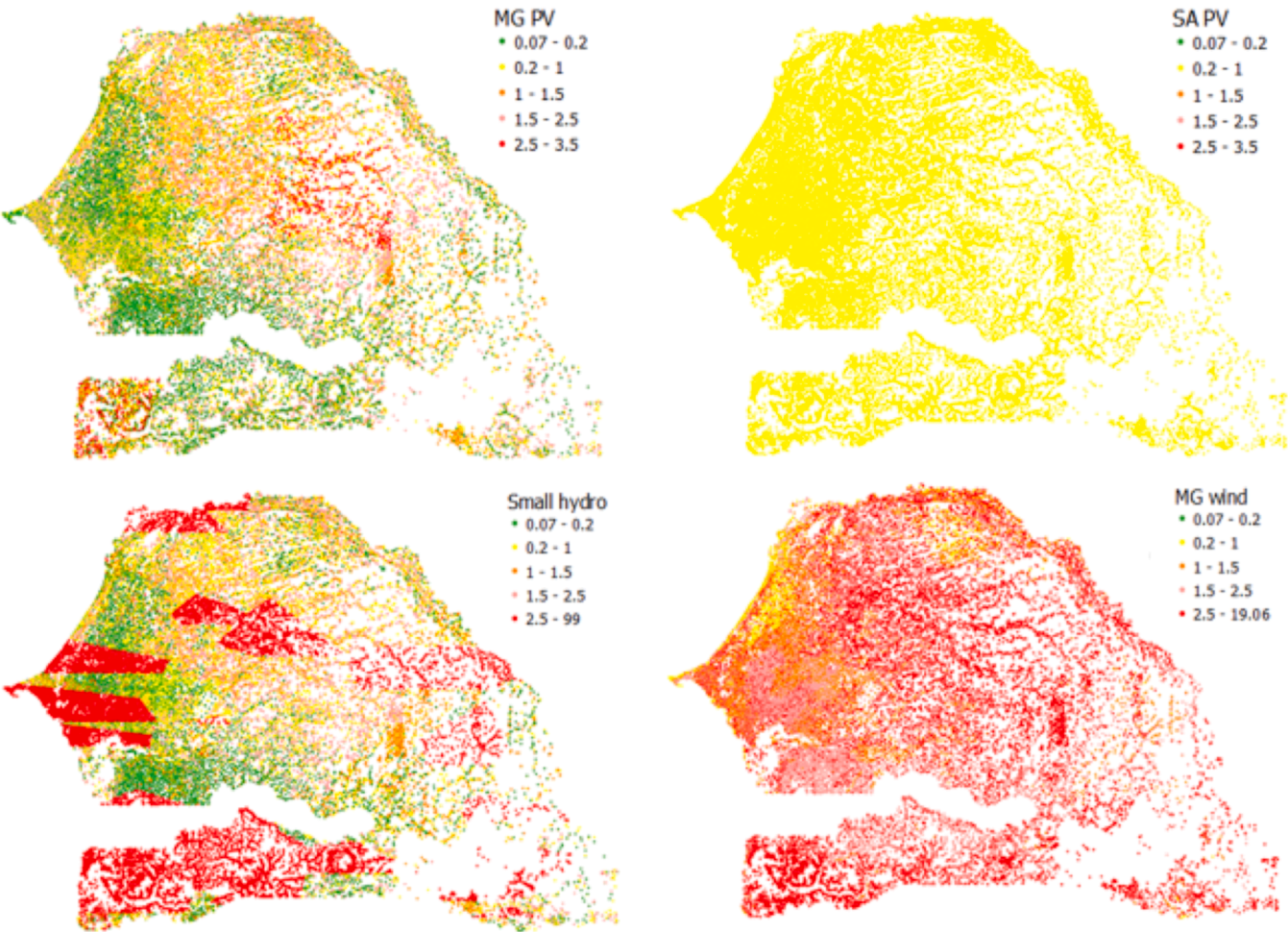


Fig. 9. LCOE estimated for off grid technology in USD/kWh in 2030.

technologies for the study area by 2030. The most convenient off-grid solutions are exclusively two: MG PV and SA PV systems.

The spatial allocation of electrification technologies in the model is primarily influenced by the geographic and socio-economic characteristics of the study area. Key factors such as solar resource availability, population density, infrastructure deployment costs, and local energy demand profiles shape the selection process. Beyond cost, the reliability of electricity supply also plays a critical role in determining the most suitable option for each locality. In settlements with moderate to high population densities, solar PV-based mini-grids often represent the most suitable option. Their shared generation and storage systems reduce unit costs and improve reliability for Tier 2 and Tier 3 demand levels. In contrast, SA PV systems are more appropriate for remote or sparsely populated areas, where grid extension or MG deployment would be economically unfeasible and technically less efficient. However, their reliability depends heavily on adequate battery sizing and system autonomy, which OnSSET accounts for through technology-specific parameters such as storage costs, depth of discharge, and system lifetime.

This distribution highlights the importance of spatial and economic considerations in off-grid electrification planning, ensuring that technology choices align with the specific constraints and opportunities present in each settlement.

Comparative analysis of the LCOE across electrification technologies

Fig. 9 presents a comparative analysis of LCOE for the different electrification technologies considered in the Senegalese case study. The LCOE, expressed in USD/kWh, is a central metric used to assess and compare the long-term economic viability of electricity supply options, taking into account capital investment, operation and maintenance costs, system lifetime, and energy output. The analysis reveals significant spatial and technological variation in LCOE outcomes, reflecting differences in resource availability, system efficiency, and local demand characteristics. The OnSSET model evaluates each settlement independently to determine the least-cost off-grid electrification strategy selecting between solar-based MG and SA systems, while also excluding technologies that do not meet economic feasibility thresholds in the local context.

- **MG PV:** The LCOE for this technology exhibits a wide range, with a minimum value of 0.07 USD/kWh and a maximum reaching 3.41 USD/kWh. On average, across all studied settlements, it stands at 0.919 USD/kWh. This variability is mainly driven by spatial differences in system costs, solar resource availability, and energy storage requirements. In the Senegalese context, the average installation cost for mini-grid solar PV systems is approximately 2580 USD/kW. Solar irradiation varies moderately between settlements, ranging from 2042 to 2159 kWh/m²/year, which directly affects the energy yield and capacity factor of PV systems at the local level. Furthermore, energy storage represents a significant share of total system costs, with battery prices assumed at 139 USD/kWh, reflecting current market values for lithium-ion technologies. These parameters together with local demand levels and settlement size explain the considerable dispersion in LCOE values across the country.
- **SA PV:** The LCOE for SA PV systems falls within a narrower range, between 0.31 USD/kWh and 0.64 USD/kWh, with an average of 0.387 USD/kWh. Although mini-grid PV systems are generally expected to benefit from economies of scale and shared infrastructure, the simulation results reveal that stand-alone PV systems exhibit a lower average LCOE (0.387 USD/kWh) compared to MG PV systems (0.919 USD/kWh). This outcome can be attributed to several inter-related factors. First, SA PV systems are predominantly allocated to low-demand, sparsely populated rural settlements, where individual household systems can be optimally sized without the need for complex distribution networks or centralized storage systems. The

simplicity of these installations reduces both capital costs and technical losses.

- **Second,** in the case of MG PV systems, higher LCOE values are often observed in small- to medium-sized settlements, where the cost of deploying a distribution network, centralized storage, and control systems is not fully offset by the number of users, especially under Tier 2 or Tier 3 demand assumptions. Additionally, the model may assign MG PV to settlements where higher autonomy or reliability constraints drive up battery storage requirements and thus increase costs.
- **Hydropower:** The hydroelectric systems assessed in this study are of very small scale, and their applicability is highly constrained by geographic and resource-specific factors. While the LCOE values for hydro range from as low as 0.11 USD/kWh up to the maximum threshold of 99 USD/kWh which in OnSSET indicates economic infeasibility the majority of settlements fall into the latter category. However, it is important to note that a limited number of settlements, located in proximity to favorable hydro sites do exhibit viable conditions for micro-hydro deployment at low cost. Nevertheless, such opportunities are rare and spatially isolated, due to topographic constraints, seasonal variability in water availability, and the relatively high unit cost of civil works and infrastructure in rural or hard-to-access areas. Consequently, hydroelectricity is only selected in very specific cases where local conditions strongly support its feasibility at minimal scale.
- **Wind Energy:** The LCOE for wind power varies between 0.30 USD/kWh and 19.06 USD/kWh, with a global average of 2.37 USD/kWh. This wide range reflects substantial variation in wind resource availability, combined with site-specific installation and maintenance costs. In the Senegalese context, wind energy is constrained by low capacity factors in most settlements typically below 10 % which severely limits annual generation. Furthermore, the intermittent nature of wind production increases system costs due to the need for additional storage or hybridization. Technical challenges related to installation, access to spare parts, and the need for specialized maintenance further raise the total cost of ownership. When compared with the more competitive and predictable performance of solar PV, wind systems are generally outperformed and are thus excluded from least-cost scenarios in the model.

These findings underscore the necessity of a thorough local assessment to determine the most cost-effective electrification solutions based on the specific conditions of each studied area.

Investment cost analysis

The total investment cost required for electrification across the studied settlements exhibits substantial variability, ranging from 233.50 USD to 4397,316.43 USD, with an overall average of 4596.24 USD. This broad range reflects the diversity of settlement characteristics and the electrification technologies assigned by the model. In small and remote settlements, where SA PV systems are typically selected, investment costs remain relatively low due to the absence of distribution infrastructure and the use of small-scale, modular systems. Conversely, in larger or more densely populated areas, the deployment of grid extension or MG solutions especially under Tier 2 service requirements involves significantly higher capital costs, including generation, storage, distribution, and connection infrastructure. The variation also incorporates geographic and spatial factors such as distance to existing grids, solar resource potential, and population density, which all influence both the scale of required investment and the optimal technology selected Fig. 10.

Table 6 presents the average investment cost per capita across different settlement population brackets. A clear inverse relationship emerges between settlement size and per capita investment cost: the smallest settlements (0–500 inhabitants) exhibit the highest average

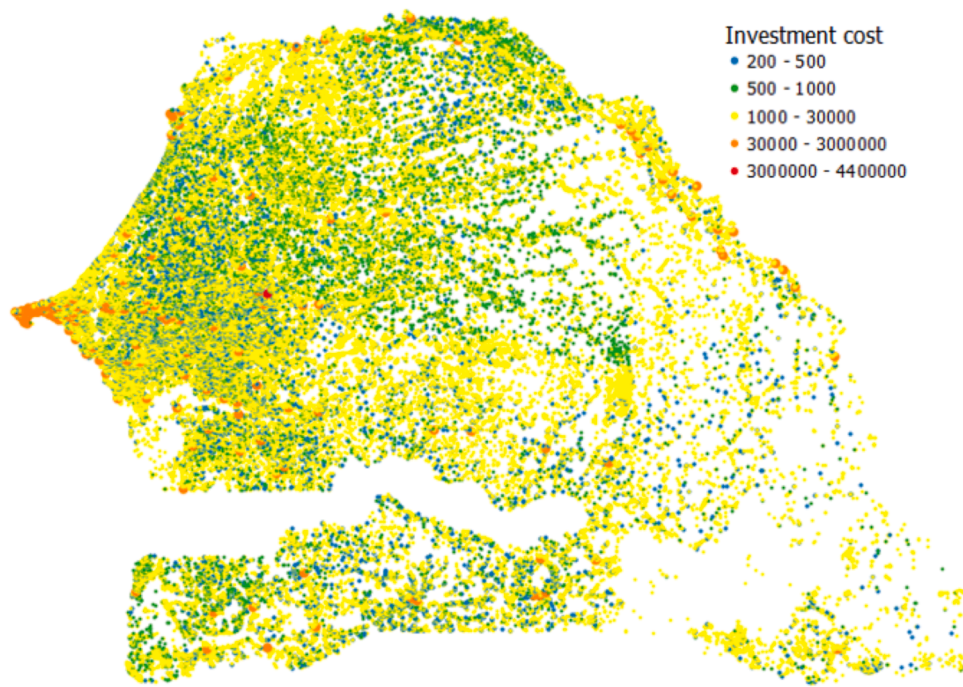


Fig. 10. Investment cost analysis.

Table 6
Investment cost per capita.

Population range (inhabitants)	Number of settlements	Average investment cost per capita (USD)
0–500	53,887	261.06
500–1000	2514	59.22
1001–5000	1960	59.10
5000–10,000	167	56.51
>10,000	173	56.25

cost at USD 261.06, while larger settlements (>10,000 inhabitants) display significantly lower costs, averaging USD 56.25 per capita. These cost estimates correspond to the least-cost technology option identified by the OnSSET model for each settlement namely grid extension, MG PV, or SA PV based on geospatial, demographic, and techno-economic parameters. The high per capita cost observed in low-population settlements is primarily due to the limited number of beneficiaries over which infrastructure and generation costs must be distributed. In contrast, larger settlements benefit from economies of scale and better infrastructure proximity, resulting in more cost-efficient electrification. These cost dynamics emphasize the importance of tailoring technology choices not only to geographic and resource conditions but also to population density and demand aggregation. This trend aligns with the technology allocation results and reinforces the importance of demographic density in least-cost electrification planning.

Regional drivers of cost variability in electrification planning

The LCOE and investment costs across Senegal is influenced by several interrelated factors. Among the most significant is the variability of solar resource availability, which impacts the energy output and performance of solar-based systems. A recent study [92] analyzing the spatio-temporal variability of solar radiation over Senegal using 14 years of satellite-derived data (Helioclim-3 V5) confirms that daily, seasonal, and interannual solar resource variability is relatively low, with a coefficient of variation (COV) ranging from 6 % to 9 % depending on the region and season. However, this moderate variability can still

cause differences in system sizing and, consequently, in LCOE estimates. Other factors contributing to cost disparities include population density, remoteness from existing grid infrastructure, terrain accessibility, and the proximity to roads and substations. These elements collectively affect technology selection and investment decisions, emphasizing the need for spatially resolved modeling in electrification planning. A study such as this, can provide important suggestions and define limitations for electricity development at national scale but they have to be verified case by case when passing to the technology deployment phase.

Analysis of electrification strategies and settlement characteristics

The number of settlements electrified by the grid in 2018 is 6682 (11.38 %). This number is used as the starting point for modeling grid extensions. This starting point of the model allows for the identification of areas that require electrification after 2018. OnSSET compares the costs of grid extension with off-grid solutions, including MG PV, MG hydro, MG wind, MG diesel, SA PV, and SA diesel.

The data presented in Table 7 and Table 8 provide a comprehensive overview of the electrification strategies implemented across different settlements in the study area. The results highlight the distribution of electrification technologies, the number of population with access to electricity, and the correlation between settlement size, power capacity, and electrification method.

Table 7

Number of settlements electrified with different technologies in the intermediate year 2027 and the final year 2030.

		Percentage
Number of settlements electrified by grid in the intermediate year	14,077	24.0 %
Number of settlements electrified by off-grid in the intermediate year	0	0 %
Total		24.0 %
Number of settlements electrified by grid in the final year	21,180	36.1 %
Number of settlements electrified by MG PV in the final year	1353	2.3 %
Number of settlements electrified by SA PV in the final year	36,168	61.6 %
Total		100 %

Table 8

Number of people with access to electricity by technology in 2030.

Technologies	Number	Percentage
Number of people with access to electricity by grid extension	20,361,442	93.7 %
Number of people with access to electricity by MG PV	163,281	0.7 %
Number of people with access to electricity by SA PV	1209,547	5.6 %

In the intermediate year (2027), only 14,077 settlements (24.0 %) can be cost-effectively electrified through grid extension, indicating that a significant proportion of rural areas will continue to lack access to modern electricity services. The value 0 for isolated systems in the intermediate year indicates that no new stand-alone systems were installed during this phase, as grid extension was the least-cost option for all settlements. By 2030, the number of settlements electrified via grid extension may increase significantly to 21,180 (36.1 %), demonstrating the potential for expansion of the national grid. MG PV systems can be implemented in 1353 settlements (2.3 %), while SA PV systems can electrify the largest number of settlements (36,168 settlements, 61.6 %), confirming that decentralized solar energy solutions are the dominant choice for rural electrification.

Despite grid extension covering only 36.1 % of settlements, 20.36 million people (93.7 %), will have a reliable access to electricity, showing that most of the high-population settlements will be connected to the grid. MG PV systems will electrify 163,281 people (0.7 %), whereas SA PV systems will serve 1.21 million people (5.6 %), reflecting their role in rural, sparsely populated communities.

Table 9 categorizes the electrification technologies used according to the different size of the settlements by population ranges in 2030. Grid extension is predominantly implemented in settlements with larger populations, particularly those with >1000 inhabitants (2300 settlements). It is also applied in smaller settlements, but with decreasing frequency as the population size decreases. MG PV systems are used exclusively in settlements with populations ranging from 100 to 500 inhabitants (1353 settlements), indicating that these systems are designed for medium-density areas where centralized solutions are not viable. SA PV systems dominate small settlements, particularly those with populations below 100 inhabitants. These systems electrify 28,209 settlements with populations below 50 and 7011 settlements in the 50–100 population range, making SA PV the primary solution for low-density rural areas.

Table 10 classifies settlements based on the required power capacity (kW) and the type of technology used. Grid extension is primarily used in settlements with higher power requirements. While a significant number of grid-connected settlements fall within the 0–3 kW range (20,629 settlements), the grid is the only technology used for settlements requiring >50 kW, highlighting its role in high-demand areas. MG PV systems are concentrated in the 5–20 kW range (1353 settlements), reinforcing their suitability for medium-scale electricity needs. SA PV systems are dominant in the lowest power range (0–3 kW), covering 31,015 settlements, aligning with their role in small, low-energy demand communities.

Several decentralized technologies were tested in the model, including diesel, wind, hydro, PV mini-grids, and SA systems. For the population sizes and demand levels considered, MG PV were consistently selected as the least-cost option. This outcome reflects Senegal's abundant solar resources, the declining costs of PV technology, and rural

Table 9

Classification of settlements by population size and electrification technology.

Population	0–50	50–100	100–250	250–500	500–1000	Above 1000	Total
Grid	5098	1514	5569	4185	2514	2300	21,180
MG PV	0	0	1353	0	0	0	1353
SA PV	28,209	7011	948	0	0	0	36,168
Total	33,307	8525	7870	4185	2514	2300	58,701

Table 10

Number of settlements by technology based on the power range (kW) 2030.

Capacity	0–3	3–5	5–20	20–50	50–100	Above 100	Total
Grid	20,629	239	154	6	58	94	21,181
MG PV	0	0	1353	0	0	0	1353
SA PV	31,015	4788	365	0	0	0	36,168
Total	51,644	5027	1872	6	58	94	58,701

load profiles that are well suited to PV-based systems.

The results indicate that settlements with higher populations have greater power demands and are more likely to be connected to the grid, while smaller settlements, characterized by lower energy consumption, rely on decentralized solutions. Settlements above 1000 inhabitants are electrified exclusively by the grid, with power requirements often exceeding 50 kW. Settlements between 100 and 500 inhabitants, which require power in the 5–20 kW range, are best suited for MG PV systems. Small settlements (below 100 inhabitants) rely almost entirely on SA PV systems, which typically require 0–3 kW of power, supporting household-level energy needs.

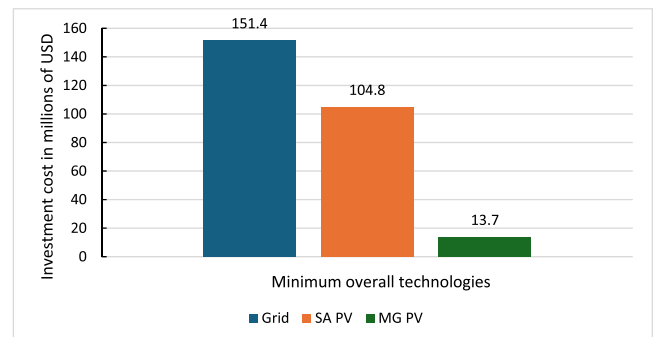
Analysis of investment costs for electrification

Fig. 11 illustrates the total investment costs required to achieve full electrification in the target year, disaggregated by technology: grid extension, SA PV systems, and MG PV systems. The total investment is determined by both the technology deployed and the share population electrified.

By the year 2030, 93.68 % of the population is expected to have access to electricity through grid extension, requiring an investment of 151.4 million USD. This high percentage reflects the predominance of grid-based electrification in densely populated areas, where infrastructure expansion is economically viable.

Meanwhile, 0.75 % of the population is electrified via MG PV systems, with an associated investment cost of 13.7 million USD. MG systems are typically deployed in settlements where grid extension is not feasible, but where population density allows for shared energy distribution, reducing per-unit costs compared to SA solutions.

Finally, 5.57 % of the population is electrified using SA PV systems, requiring a total investment of 104.8 million USD. The high cost associated with SA PV reflects the individualized nature of these systems,

**Fig. 11.** Total investment cost if electrification is achieved in 2030.

which are primarily used in remote and sparsely populated areas where neither grid extension nor MG are economically viable.

It is important to note that 269 million USD are <1 % of the Senegalese 2023 GDP and distributed in 5 years from now to 2030 is a very small investment compared to the benefits it may provide to the growth of the country.

Sensitivity analysis on technology cost assumptions

To assess the influence of cost assumptions on the selection of least-cost electrification technologies, we conducted a sensitivity analysis by varying the capital cost of SA PV systems. This was implemented through the `PV_panel_cost_factor` parameter, which was set at 0.6, 0.8, 1.0 (base case), and 1.2 to simulate cost reductions or increases. The results, illustrated in Fig. 12, show that when PV costs are reduced by 40 % (factor = 0.6), the share of the population served by SA PV increases from 5.6 % to 13.8 %. Conversely, when costs are increased by 20 % (factor = 1.2), this share falls to 3.5 %. These findings confirm that the choice of least-cost technologies is highly sensitive to the assumed costs, particularly for solar PV systems. This underlines the importance of including cost uncertainty in electrification planning, especially for remote and sparsely populated areas where decentralized PV may become increasingly competitive as prices decline.

Influence of battery storage on solar-based systems

Energy storage plays a critical role in the viability and performance of solar-based electrification solutions, particularly for mini-grids and stand-alone PV systems that must operate independently of the national grid. Batteries enable these systems to meet energy demand consistently despite the intermittent nature of solar generation, especially during nighttime hours or periods of low irradiance. In the OnSSET model, battery storage is implicitly integrated into the LCOE calculations for these technologies, affecting both investment and operational costs. This study adopts a benchmark battery cost of 139 USD/kWh, in line with recent estimates from [82]. This value reflects the declining trends in battery prices, especially for lithium-ion technologies, which are commonly considered in rural electrification contexts due to their efficiency, energy density, and decreasing costs. The system sizing, including battery capacity, is determined based on daily consumption needs, load profiles, and autonomy requirements, ensuring that the modeled systems can reliably meet Tier 2 and Tier 3 demand levels as defined by the Multi-Tier Framework.

Policy alignment and implementation constraints in the Senegalese context

The OnSSET model identifies least-cost electrification technologies. However, their deployment in Senegal depends on country-specific non-technical factors. MG and SA PV systems, although economically

optimal, require supportive regulatory frameworks, financing mechanisms, and local operational capacity. Senegal has made progress in this regard through the National Rural Electrification Action Plan and the Senegalese Rural Electrification Agency (ASER). These institutions facilitate decentralized electrification and encourage private sector participation.

National initiatives, such as Scaling Solar Senegal, the Renewable Energy Integration Plan, and public-private partnerships, demonstrate Senegal's commitment to a diversified, renewable-based electrification strategy [93]. They directly support the economic viability of solar PV in rural and peri-urban areas. Remaining challenges include the political dominance of grid extension, land acquisition issues, and limited local maintenance capacity. To address these, the government promotes local PV assembly, quality assurance, and community education programs. These measures enhance system sustainability, reduce operational costs, and improve alignment between planning and implementation.

The successful realization of cost-optimal pathways will depend on ongoing improvements in governance, financing, and community engagement. Senegal's renewable energy policies and institutional structures provide a strong foundation for this effort.

Comparative insights from other Sub-Saharan electrification studies

To contextualize the findings of this study, it is helpful to compare Senegal's least-cost electrification pathways with similar efforts conducted in other SSA countries using the OnSSET model or related geo-spatial tools. For example, studies in countries such as Ethiopia [85], Nigeria [94], and Tanzania [60] have shown that a mix of centralized grid extension and decentralized renewable energy systems is typically required to achieve universal access efficiently. Like Senegal, peri-urban areas in these countries tend to be best served by grid extension due to higher population densities and proximity to existing infrastructure, while remote and sparsely populated settlements are more cost-effectively electrified through solar MG or SA systems. However, regional differences in population distribution, solar resource potential, infrastructure coverage, and institutional readiness lead to varying levels of decentralization across contexts. In Senegal, the strong solar potential and relatively dense peri-urban clusters result in a high share of grid-extended settlements (93.68 %), which contrasts with countries like Ethiopia, where MG and SA systems play a more prominent role. These comparisons emphasize the need to adapt electrification planning to country-specific demographic, geographic, and infrastructural conditions, and to promote studies and research activities in this field, while leveraging lessons learned from neighboring Sub-Saharan countries to inform more robust policy and investment strategies.

Conclusion

This study maps least-cost electrification pathways for Senegal by 2030 using the OnSSET model. High-resolution data on population, infrastructure, and renewable resources were integrated. Results show that 93.7 % of the population can be served by grid extension, 0.7 % by solar mini-grids, and 5.6 % by stand-alone PV systems. The total investment needed is about USD 269.8 million for 116.1 MW of additional capacity. These pathways can guide SENELEC, ASER, and the Ministry of Petroleum and Energy in prioritizing investments and coordinating programs under PANER and the Renewable Energy Integration Plan. Realistic financing, including public-private partnerships and concessional loans, is essential to attract investment and reduce upfront costs. The OnSSET framework provides a transparent, scalable tool for sustainable and inclusive electrification in Senegal and similar Sub-Saharan contexts.

CRedit authorship contribution statement

Adama Sarr: Writing – original draft, Visualization, Validation,

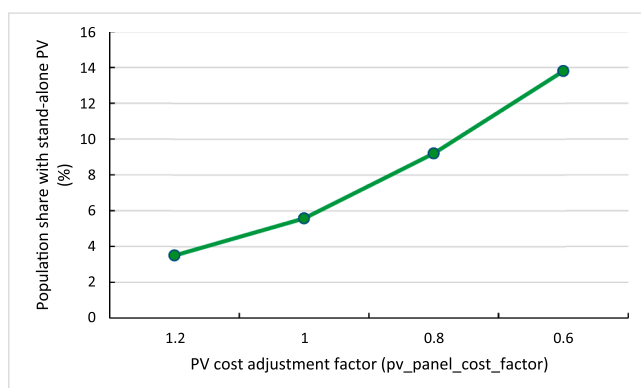


Fig. 12. Impact of PV cost variation on electrification pathways.

Software, Methodology, Data curation, Conceptualization. **Aldo Bischi:** Writing – review & editing, Validation, Methodology. **Umberto Desideri:** Writing – review & editing, Supervision. **Cheikh Mouhamed Fadel Kebe:** Writing – review & editing, Supervision.

Declaration of competing interest

We have no conflicts of interest to declare.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.nexus.2025.100621](https://doi.org/10.1016/j.nexus.2025.100621).

Data availability

Data will be made available on request.

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